Modules – Boards

Menu: Modules > Boards

Lists modules 0–7 and every board RateController can hear. This is where you choose the module the module pages edit, give a board its module number and description, and load a starting configuration. Each board is known by its built-in hardware (MAC) number, so boards are told apart even when they are the same type or share a module number. The list updates every second.

Board List

Every module 0–7 has a row, with or without a board. New boards are listed last. The rows of the module being edited are shown in **bold**.

Column	Description
Module	The module number, 0–7, or **New** for a board that has no module yet.
Board	The board type, the last part of its MAC number, and its description, e.g. *ESP Rate ...0B:B0:12 Left Tank*. A dash when the module has no board.
State	**Connected** — the board is reporting. **No board** — no board is reporting on this module. Its settings can still be edited. **Duplicate** — two or more boards are on the same module. Each is listed on its own row. Move one of them to a free module. **Update firmware** — a board on this module has firmware too old for this version of RateController. The module is not used until the firmware is updated. **New** — a new or blank board waiting for a module. **→ 3, not sent** or **Not sent** — a saved change that goes out with the next Send. The row of the module a board is moving to shows the same.

Choosing the Module to Edit

Click a row to choose that module. The Config, Communication, Pins, Relay Pins and Valves pages, and the Send button, then work on it. The choice takes effect straight away, without Save, and the row turns bold.

- A row for a module with no board can be chosen too, for example to set a module up before its board is connected.
- A board with a saved, not yet sent, module change chooses the module it is moving to.
- A **New** board doesn't change the module being edited, because it has no module yet.
- Choosing another module is refused while a module page has unsaved changes. Press Save or Cancel on that page first.

> **Note:** Choosing a row, or moving a board to another module, does not change any settings — neither RateController's settings for the module nor the board's own. It only changes which module's saved settings the module pages show. To change them, use **Load Presets** or enter them on the Config, Pins, Relay Pins and Valves pages, Save, then press Send.

Settings

These change the selected board. They can't be used on a row with no board.

Setting	Description
Module	The module the board serves. The list holds the board's own module and the free modules only: a module that already has a board, including one that needs a firmware update, is not offered.
Description	A name for the board, up to 16 characters, e.g. "Left Tank". It is stored on the board and kept through firmware updates. The box can't be used for a Nano (it stores no description), for a board on a duplicate module, or for a new board until a module is chosen.
Web page link	For an ESP32 on WiFi, the address of its web page. Click it to open the page. An ESP32 on Ethernet shows *Ethernet, no web page*.

Load Presets

Load Presets loads a full starting configuration for a known board into the module being edited — the settings on the Config, Pins and Relay Pins pages — and saves it.

- On a row with a board, it asks to confirm the preset for that board's type.
- On a row with no board, it asks which type to load, then shows a message that it was saved.

Preset	Microcontroller
Nano (RC12-3)	Arduino Nano
Teensy (RC11-2)	Teensy 4.1
ESP32 (RC15)	ESP32

After loading, check the Pins, Relay Pins and Valves pages against your own board, then press **Send**. Send only lights up when a board is connected on the module; a preset loaded for a module with no board is kept until a board is set up there.

- Load Presets is refused while this page or a module page has unsaved changes.
- It can't be used with a **New** board selected until the board has a module.
- A preset replaces the module's pin and relay settings. Only use it during initial setup or when changing hardware. It does not change the module number.

Setting Up a New Board

New and blank boards start without a module (ID 25). While one is reporting, the **Boards** button on the menu has an orange border.

One new board. When RateController hears one new board and no other boards, it asks "*New board found: ESP Rate ...0B:B0:12. Set it up as module 0?*" Press **Yes**. The board gets module 0 and the starting configuration for its type, then restarts and reconnects as module 0. Check the Config, Pins, Relay Pins and Valves pages against your own board afterwards.

- The question is asked once per board each time RateController runs. If you answer No, set the board up from this page instead.
- It isn't asked when other boards are connected, when two or more new boards are reporting, or while a module page has unsaved changes.

Several boards, or a module other than 0:

1. Connect the board. An ESP32 on WiFi is set up from **Modules > Communication > WiFi** instead, which gives it its module at the same time.
2. Open **Modules > Boards** and select the board marked **New**.
3. Choose a free **Module**, add a **Description** if you like, and press **Save**. The module pages switch to the chosen module.
4. RateController offers the starting configuration for the board's type (Nano, Teensy or ESP32). Press **OK** to load it, then check the Config, Pins, Relay Pins and Valves pages against your own board.
5. Press **Send** ⬆ . The board takes its module number and the module's settings, restarts, and reconnects as that module.

Set up two new boards one at a time: Send the first and wait for it to reconnect before saving the second.

A New Board on Another Network

A new board on Ethernet starts on network 192.168.1. RateController hears it even when this PC is on a different network, and the State column shows **New, network 192.168.1**. When it's set up, with the question above or with Send, the board is also moved to RateController's module network.

- If this PC also has a connection on the board's network, RateController asks "*Move the board to network 192.168.10? It is on 192.168.1.*" Answer **No** to keep it where it is, then set RateController's module network to match on **Modules > Communication**.
- Otherwise the board is moved without asking, because RateController couldn't reach it on its old network.
- An ESP32 set up over its own hotspot isn't asked. Its hotspot is renamed with its new module (e.g. *RateModule_0_...*) and the PC joins the new name by itself. For a hotspot with a password, a message says which hotspot to join.

Moving a Board or Changing its Description

1. Select the board, choose the new **Module** and/or type a new **Description**, and press **Save**. The State column shows the change as *not sent*. A module change also moves the module pages to the new module.
2. Press **Send**. A board moved to a new module gets only its new module number (and its description, if changed). It keeps its own settings, restarts, and reconnects on that module. When only descriptions have changed, Send sends just the descriptions and the boards don't restart.

- RateController's saved settings for the new module are not sent with the move, and may not match the board. Check them on the module pages. To send them, press Save on a module page, or use **Load Presets**, then press Send once the board has reconnected.
- Module page changes saved before the move are not sent with it. Send stays lit, and the next Send gives them to the board on its new module.
- Only one module change can wait for Send at a time. Descriptions can be saved for any number of boards.
- To undo a saved module change, select the board's own module again and press Save. The module pages go back to the board's module.
- Another row can't be selected while the current board has unsaved edits. Press Save or Cancel first.

When Send is Refused

Send does not go out, and a message says why, when:

- the board being moved has stopped reporting, or another board has appeared on its new module;
- the module being edited has no board connected, or its board needs a firmware update (only when module settings are being sent; a description-only Send doesn't check this);
- a module page has unsaved changes after a module change ("*Save or cancel the pending changes first.*"). Save or cancel them and press Send again.

Except for the last case, saved module changes and descriptions are dropped when Send is refused, so they can't go out later by surprise. Save them again when the board is ready.

> **Note:** A Nano has no description. It is shown by its MAC number.

> **Note:** A board that is switched off isn't seen, so its module looks free. Make sure every board is powered before moving boards around.

> **Note:** A board with older firmware may not appear here at all. An older Nano, for example, isn't listed and its module shows as *No board*. Update the board's firmware, then it appears as its module or as New. See **Updating Module Firmware** on the Help page.

Controls

Control	Action
Load Presets	Load and save the starting configuration for the module being edited (see above)
Cancel	Discard the changes to the selected board
Save	Hold the changes until the next Send
Send	Send saved board changes, then the settings of the module being edited. A board moved from one module to another gets only its new module number